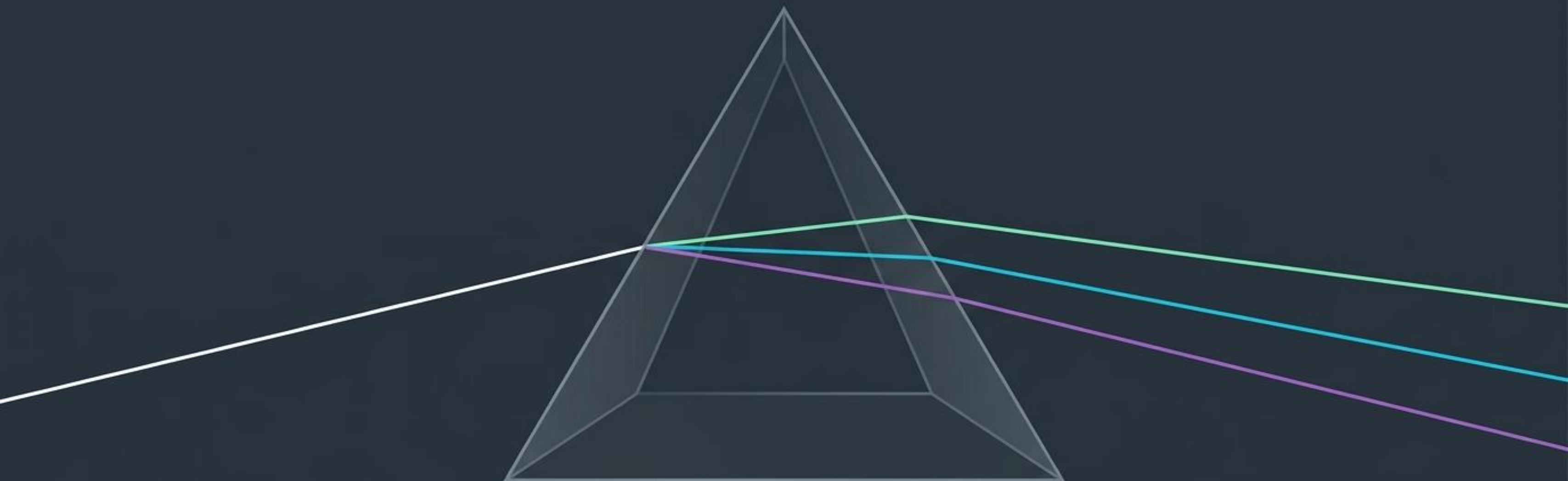




Enterprise Knowledge Intelligence Platform

Hybrid Relational, Document, and Vector Data Architecture

Project Team:

- M. Sathya Krishna - 2510080006
- Yashwanth - 2510080001
- Aneeq - 2510080005
- Abhinav - 2510080002

Guide: Dr. P. Anitha

Department of Artificial Intelligence and Data Science

KL University (Koneru Lakshmaiah Education Foundation)

Enterprise knowledge remains trapped by rigid search barriers

Organizations run on documents, but critical knowledge is frequently inaccessible due to three fundamental limitations of legacy systems:



01

1. The Exact-Match Constraint

Users must know the precise wording to find a document. A query for "rules for working from home" completely misses a policy titled "Hybrid and Remote Working Standard."

02

2. The Typo Tolerance Failure

A simple typing mistake like "remte working standrd" yields zero results, frustrating users and obscuring available data.

03

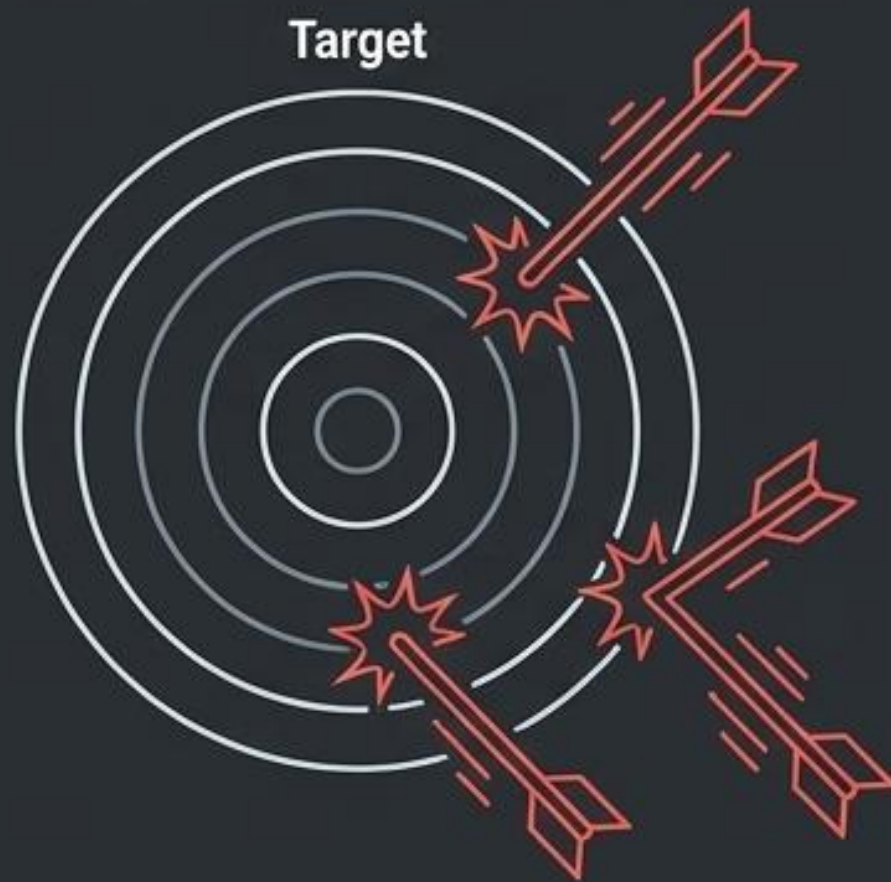
3. The Security Silo Dilemma

Information must be tightly controlled. A search engine that returns a restricted title, or includes it in a total result count, constitutes a security leak.

Solving meaning, error tolerance, and strict document access

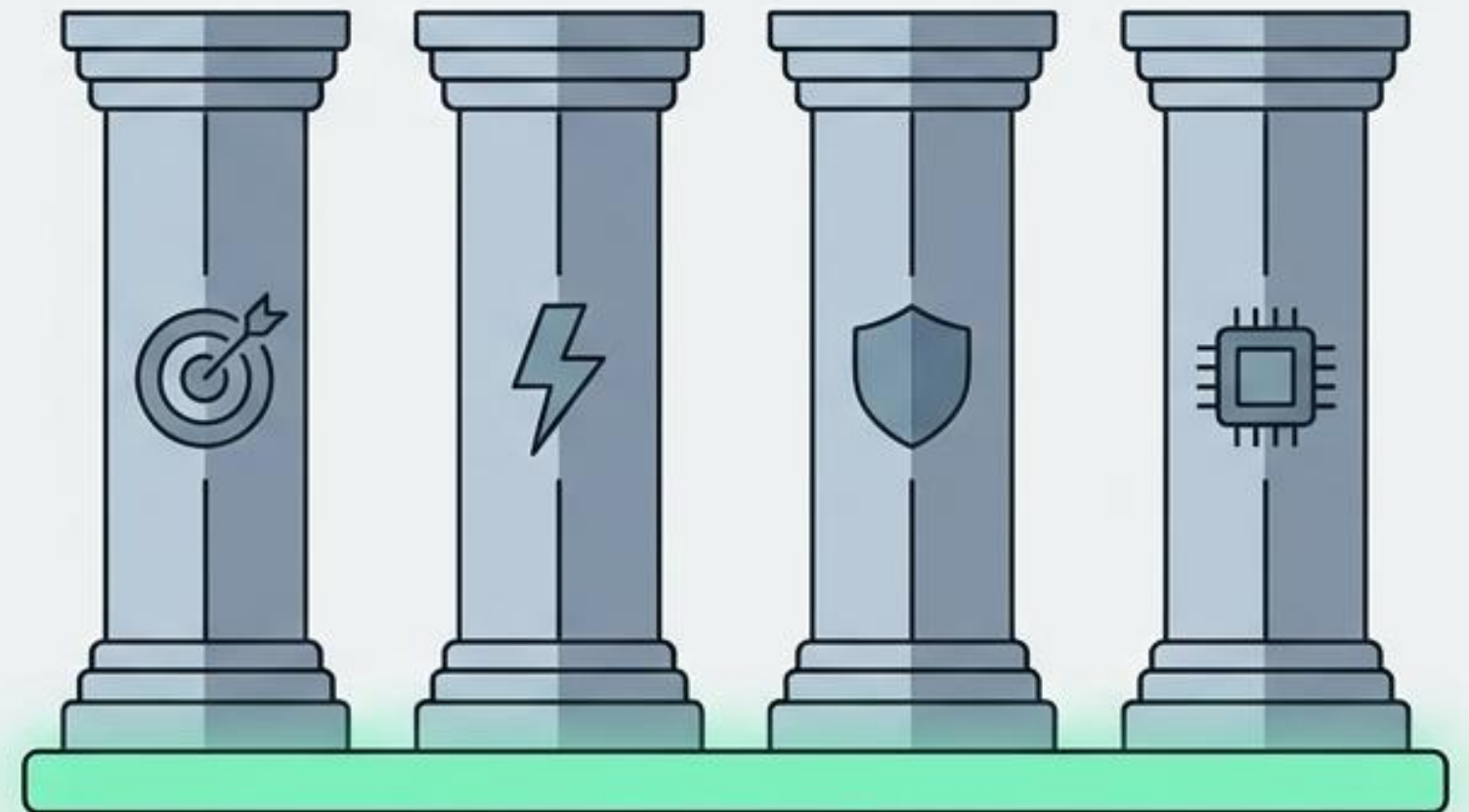
Problem Statement

Legacy enterprise search fails to understand the meaning behind queries, breaks entirely on minor input errors, and lacks the capacity to enforce granular, document-level security within the search ranking itself.



Objectives & Expected Outcomes

- **Hybrid Search Engine:** Combine lexical (exact/typo) and semantic (meaning) search in a unified request.
- **Polyglot Consistency:** Guarantee synchronization across Relational, Document, and Vector data stores. And. deploy and execute flawlessly within a strict 512MB memory limit.



Existing systems fail to balance accuracy with strict access control

[Dimension]	Legacy SQL Search	Basic Vector DBs	Standard RBAC	The EKIP Solution
Search Accuracy	Exact match only	✗ Understands meaning	✗ N/A	✗ Hybrid (Semantic + Lexical) ✓
Typo Tolerance	Breaks on error	✗ N/A	✗ N/A	✗ Algorithms (Edit distance) ✓
Access Control	No row-level security	✗ No precise joins	✗ Role-based only	✗ Granular Per-Document ✓
System Output	Data tables	✗ Distance scores	✗ Access lists	✗ Unified Intelligence ✓

Overview of the Proposed Solution

EKIP is a multi-disciplinary enterprise product that fuses relational data, advanced algorithms, machine learning, and OS-level memory management into a singular platform.

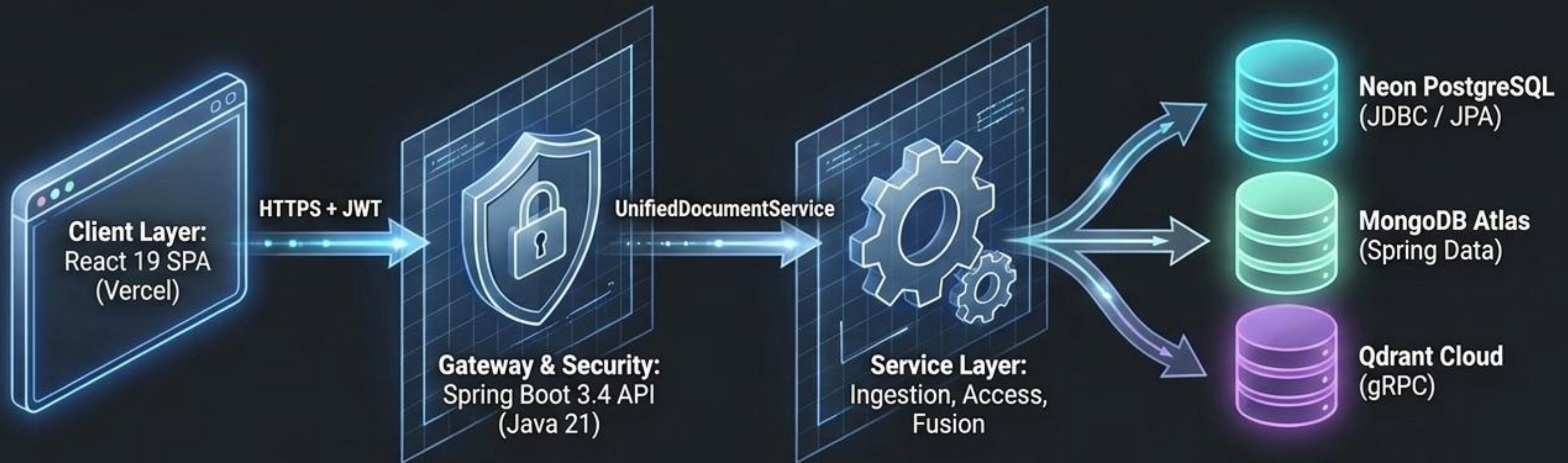
Four computer science disciplines converge into
into one intelligence platform



Key Features & Integration

- **Database Engineering:** Polyglot persistence across PostgreSQL, MongoDB, and Qdrant.
- **Machine Learning:** On-the-fly text chunking (180 words, 40 overlap) and embedded MiniLM ONNX processing.
- **Advanced Algorithms:** Custom TextHack engine powering a typo-tolerant hybrid fusion search (0.4 keyword + 0.6 vector).
- **Systems Programming:** Strict memory tuning, MALLOC_ARENA_MAX=2 caps, and single-thread inference to fit a massive workload into 512MB RAM.

A single-origin API distributing work across a polyglot data layer



1. Client Layer: React 19 SPA served by Vercel automatically proxies `/api/*` requests to bypass CORS.

2. Gateway & Security: Requests enter the Spring Boot API where JWT validation and permission filters are applied.

3. Service Layer: The `UnifiedDocumentService` coordinates Document Ingestion, Access, and Search Fusion.

4. Database Layer: Parallel queries fire to PostgreSQL, MongoDB, and Qdrant.

Three specialized databases linked by a single immutable identifier

Polyglot Matrix

PostgreSQL (Metadata & Security)



What it Holds: Users, roles, 12-table 3NF schema, document metadata, search history.

Why this Store: Enforces relational integrity, foreign keys, and absolute transactional certainty for RBAC permissions.

MongoDB (Document Content)



`postgres_document_id`

What it Holds: Extracted raw text, chunk arrays, and flexible metadata via \$jsonSchema validators.

Why this Store: Optimized for long, variable-shaped text and source processing details that break rigid SQL schemas.

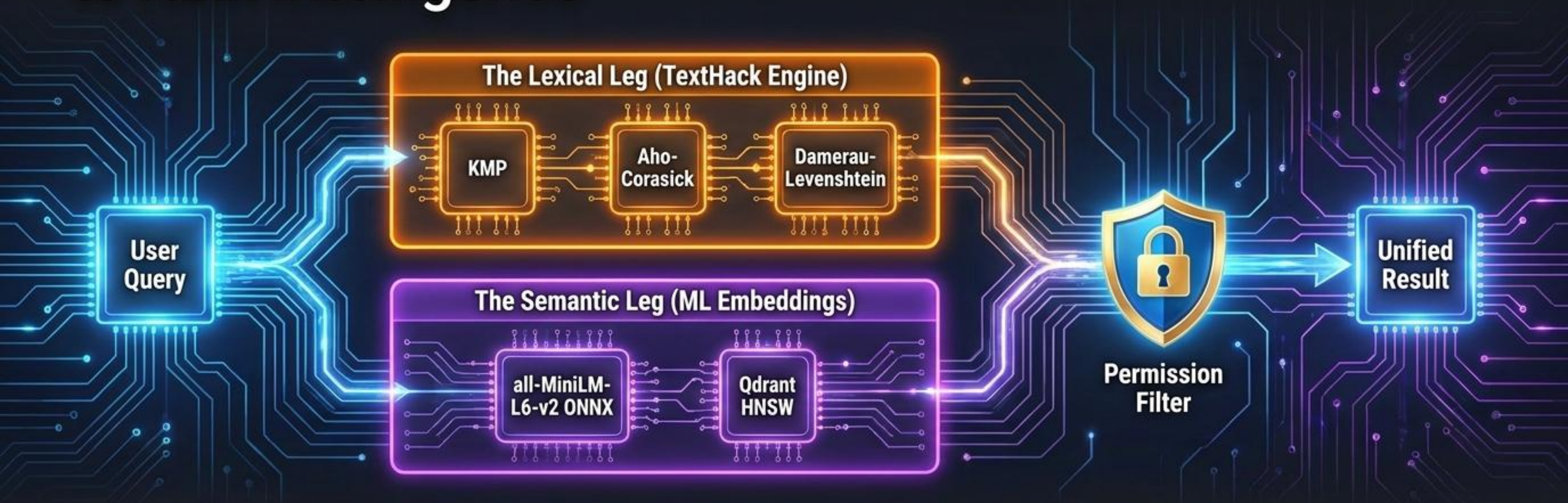
Qdrant (Semantic Meaning)



What it Holds: One 384-dimensional cosine vector per text chunk, carrying a secure payload.

Why this Store: Enables approximate nearest-neighbor search by semantic meaning (HNSW).

Fusing string algorithms with machine learning to rank intelligence



1. The Lexical Leg

- **KMP Algorithm:** Exact phrase matches in linear time.
- **Aho-Corasick Automaton:** Checks all query terms in a single pass (15.5x faster than repeated KMP).
- **Damerau-Levenshtein:** Scores fuzzy matches and transpositions (e.g., recieve → receive).

2. The Semantic Leg

- Converts query into a 384-dimensional vector.
- Executes a fast nearest-neighbor lookup in Qdrant.

3. Fusion & Normalization

- **Fused Score** = $(0.4 * \text{Keyword Score}) + (0.6 * \text{Semantic Cosine})$
- Cosine mathematically mapped from $[-1, 1]$ to $[0, 1]$ using $(\cos+1)/2$

A modern, high-performance stack built for production deployment



Ensuring polyglot data integrity via compensating transactions

Implementation: The Upload Lifecycle

To maintain consistency across three databases without distributed locks, EKIP uses the Saga Pattern (Compensating Transactions).



The Rollback Mechanism: If Step 6 fails, reverse compensating actions execute. MongoDB document is deleted, then PostgreSQL row is deleted, leaving zero orphaned data.

Sub-second semantic retrieval with enterprise-grade visualization



Output Screens & Modules

- **Primary Dashboard:** Live system metrics, recent documents, and drag-and-drop ingestion.
- **ML Insights Page:** Visualizes document classification accuracy and t-SNE clustering (k=24 clusters mapped to 21 distinct document topics).
- **TextHack Workbench:** Exposes underlying algorithms for live pattern search and complexity analysis.

Performance Observations (Render Free Tier)

- **Semantic Search Query Time:** 0.4 to 0.9 seconds (down from 9.4s after aggressive memory tuning).
- **Hybrid Search Query Time:** 0.5 to 0.6 seconds.
- **Document Extraction:** Parses a 20-page, 5,000-word .docx file into 36 vector chunks in 1.2 seconds.

Rigorously validated for scale, security, and mathematical parity

191 & 148

Backend & Frontend Tests

169 Req/s

0 Errors at 1GB Load

100%

Vector Parity (Java vs Python)

Test Suites & Coverage

- **Backend:** 191 automated tests (JUnit 5, MockMvc).
- **Frontend:** 148 automated tests (Vitest + Testing Library).
- **Systems (ShellForge):** 148 C tests, passing AddressSanitizer and Valgrind with 0 memory leaks (58 allocations / 58 frees).

Defects Caught & Fixed

- **Lock Contention Bottleneck:** Rebuilding the JWT parser per request caused massive queuing.
- **Solution:** Moved to startup singleton, raising throughput from 149 to 181 req/s.

Securing and surfacing knowledge across the modern enterprise

Benefits of the Proposed System

EKIP finds documents users didn't know existed, handles real-world typing errors gracefully, and secures sensitive data without exposing result counts or metadata.



Human Resources (HR Portals)

Benefit: Employees searching for 'vacation rules' instantly find the 'Annual Leave Policy' through semantic meaning, bypassing the need for exact terminology.



Information Technology (Runbooks)

Benefit: IT staff pasting error logs with typos find the exact system recovery document instantly, powered by Damerau-Levenshtein edit-distance algorithms.



Legal & Finance (Contract Discovery)

Benefit: Sensitive financial reports are completely hidden from standard employees via strict PostgreSQL row-level grants, preventing accidental data leaks.

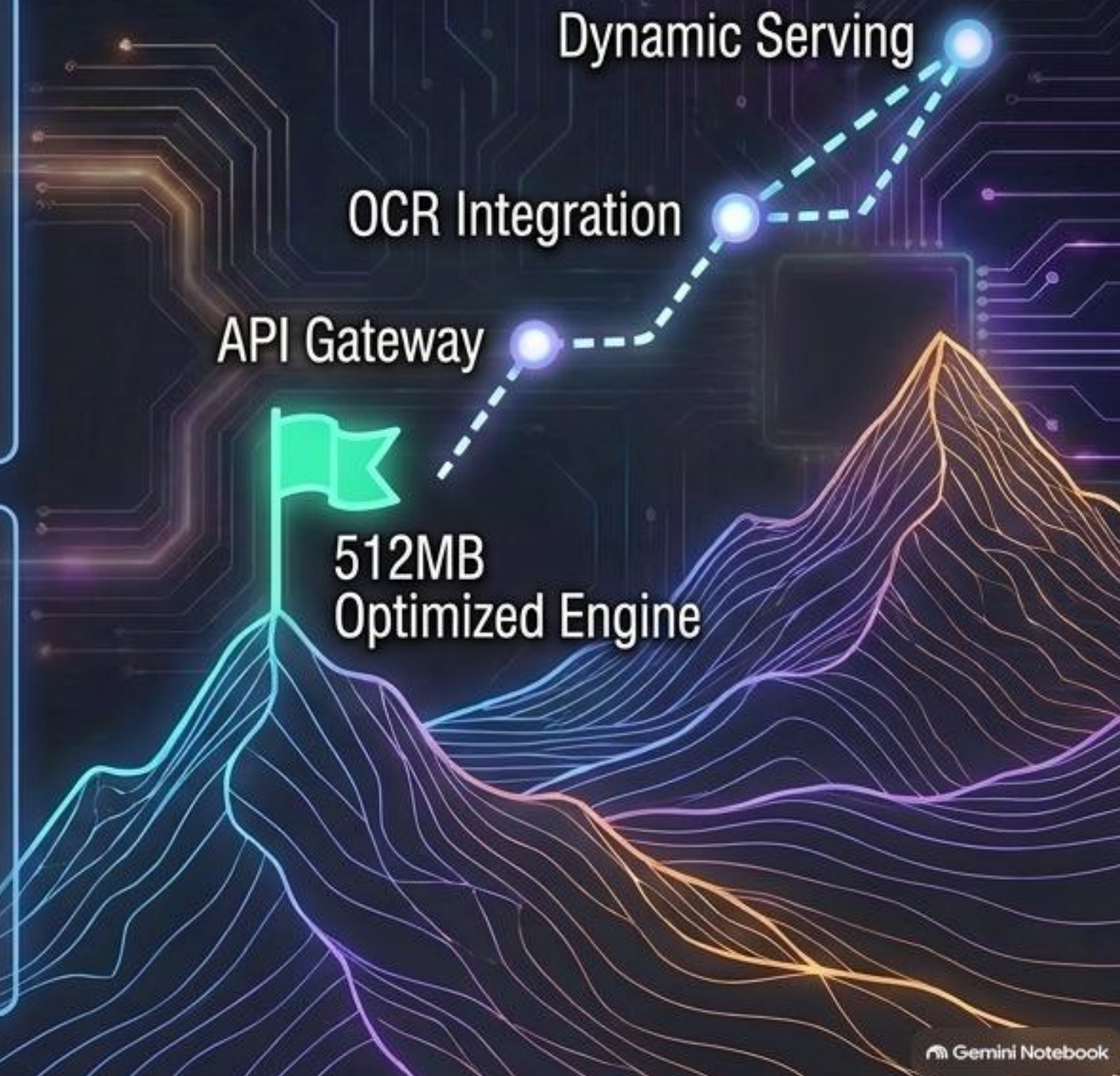
Overcoming the 512MB memory wall and mapping the future

Primary Challenge: The Render Free Tier Limit

- **Issue:** The backend (Java + ONNX model) initially crashed under a strict 512MB RAM limit. ONNX Runtime consumed excessive CPU by spawning threads per core.
- **Solution:** Severely constrained the JVM (128MB heap), limited `MALLOC_ARENA_MAX=2`, and restricted ONNX to a single inference thread. The system now runs perfectly at 446 MiB steady state.

Future Enhancements (Scope for Improvement)

- **Microservices Migration:** Split ingestion and search behind an API Gateway.
- **Optical Character Recognition (OCR):** Integrate OCR pipelines to extract text from scanned, image-based PDFs.
- **Dynamic ML Serving:** Utilize the trained Naive Bayes classifier to automatically suggest document categories upon upload.



The successful convergence of data, meaning, and security

Summary of Achievements

The Enterprise Knowledge Intelligence Platform successfully unifies complex relational modeling, distributed systems architecture, string algorithms, and machine learning into a single, production-ready product. By replacing rigid keyword barriers with semantic understanding and typo-tolerance—while guaranteeing absolute data security—EKIP transforms scattered enterprise files into secure, accessible intelligence.

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